A

3. This is a systematic random sample, considered to be reasonably unbiased.

4. Options (i), (ii) and (iv)

5. (a) III (b) I (c) V (d) IV (e) II

6. (a) Neither is there matching of individuals across the two groups, nor is there before – after comparisons on the performance of individuals. This is not a matched pairs study.

   (b) Were employees self-selecting the procedure (old or new) they used? If so, the study is observational. If they were assigned procedures to use, then it is an experiment.

7. (a) iii (b) ii (c) iv (d) i

8. (a) unimodal, right-skewed

   (b) mean

   (c) 5, 8.5, 13, 19, 22.5
   $22.5 - 5 = 17.5$
   $19 - 8.5 = 10.5$
   } all of these are estimates

   (d) $q_3 + (1.5)(\text{iqr}) = 19 + (1.5)(10.5) = 34.75$

9. $\bar{x} = \frac{1}{4}(23 + 16 + 17 + 20) = \frac{1}{4}(76) = 19$

   $\text{Variance} = \frac{1}{3}\left[(23-19)^2 + (16-19)^2 + (17-19)^2 + (20-19)^2\right]$

   $= \frac{1}{3}(16 + 9 + 4 + 1) = \frac{1}{3}(30) = 10$

   $\Rightarrow \quad s = \sqrt{\text{variance}} = \sqrt{10} \doteq 3.1623$

10. (a) $P(X=4) = 1 - (0.11 + 0.15 + 0.38 + 0.26) = 0.1$

    (b) $P(X \leq 1) = P(X=0) + P(X=1) = 0.11 + 0.15 = 0.26$

    (c) $P(X=1 \mid X \leq 2) = \frac{P(X=1 \text{ and } X \leq 2)}{P(X \leq 2)} = \frac{P(X=1)}{0.11 + 0.15 + 0.38} = \frac{0.15}{0.64} = 0.2344$

    (d) $\mu_X = (0)(0.11) + (1)(0.15) + (2)(0.38) + (3)(0.26) + (4)(0.1)$

    $= 0 + 0.15 + 0.76 + 0.78 + 0.4 = 2.09$

11. $P(\text{Closed}) = P(\text{Snows and Closed})$

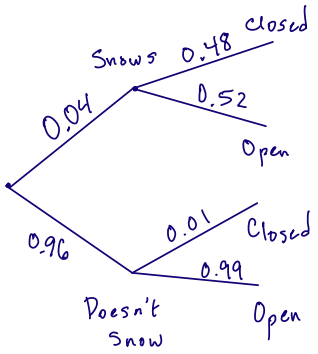
$+ P(\text{Doesn't Snow and Closed})$

$= P(\text{Snows})\, P(\text{Closed} \mid \text{Snows})$

$+ P(\text{Doesn't Snow})\, P(\text{Closed} \mid \text{Doesn't Snow})$

$= (0.04)(0.48) + (0.96)(0.01)$

$= 0.0192 + 0.0096 = 0.0288$



12. Options (a), (c), (d) and (f)

13. (a) Willa's count X is binomial, as each time she draws another token, the probability of success (green) has been restored to its original value.

(b) $n = 6, \quad p = \frac{7}{19}$